

LUC Blassel

Researcher - Machine Learning for Biology and Evolution

✉ luc@lucblassel.com 🏠 lucblassel.com

📍 Paris, France

Experience

Post-Doctoral Researcher, [Sorbonne Université](#), Paris, France 2023–
Likelihood-free inference of phylogenetic trees within the [PhyloFormer](#) project. Work with [Laurent Jacob](#) at the CQSB.

Doctoral candidate, [Institut Pasteur](#), Paris, France 2020–2023
PhD in computational biology, focused on improving and learning from sequence alignments.
Funded by [PRAIRIE](#) and supervised by [Rayan Chikhi](#). Manuscript [here](#).

Research Engineer, [Institut Pasteur](#), Paris, France 2018–2020
Using machine learning and large datasets to explore the drug resistance landscape in HIV.
Supervised by [Olivier Gascuel](#). Project page [here](#).

Data Scientist Intern, [CapitalData](#), Paris, France Mar-Sept 2017
Development of automated data validation and anomaly detection pipelines using R, Golang and Python.

Education

Doctorate (PhD), [Sorbonne Université](#), Paris, France 2020–2022
Worked at the Institut Pasteur on machine learning techniques applied to biological sequence alignments.

Masters of Science (MSc), [Dauphine université](#), Paris, France 2017–2018
[IASD](#) program, courses in Deep learning, reinforcement learning and other machine learning techniques.

Diplôme d'Ingénieur, [AgroParisTech](#), Paris, France 2014–2018
Multi-disciplinary curriculum with courses in biology, statistics, and compute science. Last year specialization in machine learning, [IODAA](#) program

BCPST Preparatory class, [École Saint-Hilaire](#), Paris, France 2012–2014
2 year French [preparatory class](#) to prepare a nationally ranked exam for entry in the "Grande Écoles"
ranked 201/1967 (Veto), ranked 481/2991 (Agro)

Publications ([Google scholar](#), [ORCID](#))

^f co-first authors

Journal articles

Phyloformer: Fast, accurate and versatile phylogenetic reconstruction with deep neural networks
[Luca Nesterenko](#)^f, **[Luc Blassel](#)**^f, [Philippe Veber](#), [Bastien Boussau](#), [Laurent Jacob](#)
Molecular Biology and Evolution, msaf051, **2025**, [10.1093/molbev/msaf051](#)

Mapping-friendly sequence reductions: Going beyond homopolymer compression
[Luc Blassel](#), [Paul Medvedev](#), [Rayan Chikhi](#)
iScience 25.11, p. 105305, **2022**, [10.1016/j.isci.2022.105305](#)

Using Machine Learning and Big Data to Explore the Drug Resistance Landscape in HIV
[Luc Blassel](#), [Anna Tostevin](#), [Christian Julian Villabona-Arenas](#), [Martine Peeters](#), [Stéphane Hué](#), [Olivier Gascuel](#)
PLOS Computational Biology 17.8, e1008873, **2021**, [10.1371/journal.pcbi.1008873](#)

Drug Resistance Mutations in HIV: New Bioinformatics Approaches and Challenges
[Luc Blassel](#)^f, [Anna Zhukova](#)^f, [Christian J Villabona-Arenas](#), [Katherine E Atkins](#), [Stéphane Hué](#), [Olivier Gascuel](#)
Current Opinion in Virology 51, pp. 56–64, **2021**, [10.1016/j.coviro.2021.09.009](#)

Origin, Evolution and Global Spread of SARS-CoV-2
[Anna Zhukova](#), **[Luc Blassel](#)**, [Frédéric Lemoine](#), [Marie Morel](#), [Jakub Voznica](#), [Olivier Gascuel](#)
Comptes Rendus. Biologies 344.1, pp. 57–75, **2021**, [10.5802/crbio.29](#)

COVID-Align: accurate online alignment of hCoV-19 genomes using a profile HMM
[Frédéric Lemoine](#), **[Luc Blassel](#)**, [Jakub Voznica](#), [Olivier Gascuel](#)
Bioinformatics (btaa871), **2020**, [10.1093/bioinformatics/btaa871](#)

Preprints & Submitted articles

Teddy: Neural Inference of Epidemiological Parameters from Viral Sequences

Vincent Garot, **Luc Blassel**, Luca Nesterenko, Anna Zhukova, Samuel Alizon, Laurent Jacob

bioRxiv, p. 2026.01.05.697728, **2026**, [10.64898/2026.01.05.697728](#)

Likelihood-free inference of phylogenetic tree posterior distributions

Luc Blassel, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

arXiv, **2025**, [10.48550/arXiv.2510.12976](#)

Conference talks & Posters

Deep end-to-end likelihood-free inference of phylogenetic trees

Luc Blassel, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

MLCB, 2025

Deep likelihood-free inference of phylogenetic trees

Luc Blassel, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

MASAMB, 2025

Likelihood-free inference of phylogenetic tree posterior distributions

Luc Blassel, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

LEGO, 2025

Likelihood-free inference of phylogenetic tree posterior distributions

Luc Blassel, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

LEGEND, 2025

Phyloformer: Fast, accurate and versatile phylogenetic reconstruction with deep neural networks

Luca Nesterenko ^f, **Luc Blassel** ^f, Philippe Veber, Bastien Boussau, Laurent Jacob

MCEB, 2024

Mapping-Friendly Sequence Reductions: Going beyond Homopolymer Compression

Luc Blassel, Paul Medvedev, Rayan Chikhi

RECOMB-SEQ, 2022

Machine learning approaches to reveal resistance mutations in HIV

Luc Blassel, Anna Zhukova, Olivier Gascuel

MCEB, 2019, (poster)